

**SASTRA UNIVERSITY
SRINIVASA RAMANUJAN CENTER, KUMBAKONAM**

ECE300 – MINI PROJECT

Project Title: LeafGuard AI: Predictive Modeling of Plant Stress Using Multisensor Data

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Abstract: Plants are highly vulnerable to environmental stresses caused by heat, drought, and humidity fluctuations, which significantly reduce growth and yield. This project introduces LeafGuard AI, a predictive framework for plant stress that employs multisensory data and machine learning methodologies. The system uses environmental and physiological factors like temperature, relative humidity (RH), vapor pressure deficit (VPD), volumetric water content (VWC), plant weight, and electrical impedance to check on the health of plants. A representative dataset was utilized, succeeded by preprocessing and feature selection to optimize model efficiency and performance. A combination of Adaboost, decision tree, and random forest, called the Hybrid-AdapTree model, was suggested to make predictions more accurate and easier to understand. The Hybrid-AdapTree model outperformed the others in experimental tests using several regression metrics. It had a R^2 score of 0.999206 for predicting impedance, a low RMSE of 62.34, and a high Pearson correlation of 0.9996. The model was able to predict temperature with a competitive level of accuracy ($R^2 \approx 0.88$). Also, feature selection makes the training process more efficient while keeping the accuracy high and lowering the amount of computation needed. The obtained results prove the reliability and efficiency of the suggested AI-powered framework in accurate processing of sensor data in order to detect plant stress.

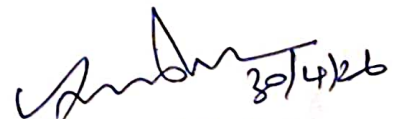
Keywords: Plant stress detection, Temperature, Humidity, vapour pressure deficit (VPD), volumetric water content (VWC), AdapTree algorithm, Machine learning.

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